AI33-MPOPT · SEVENTH PILLAR · 2026

The Cosmic Web Atlas

A Geometric Mapping, Inception to Supercluster

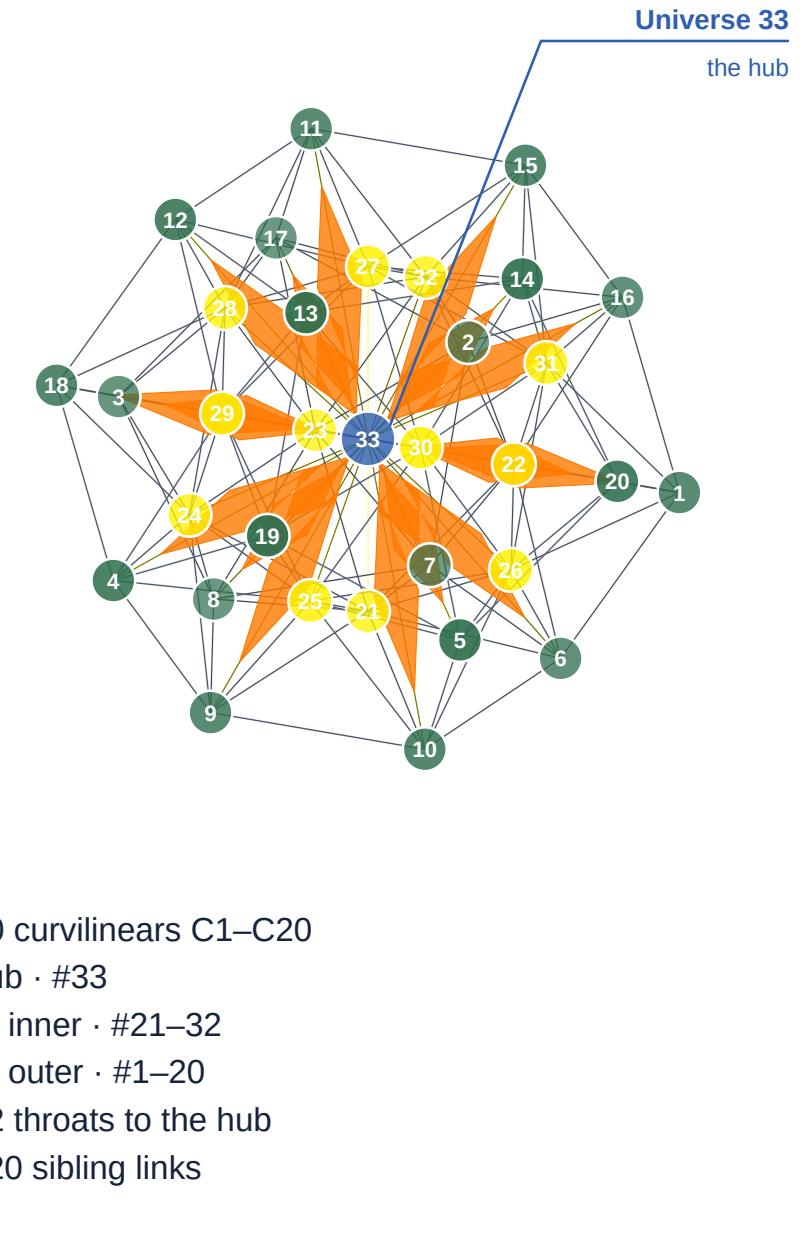
One architecture, followed through every scale it reaches — from the moment the throats open to the filaments and voids of the observed universe.

WHAT THIS ATLAS SHOWS

The 33-domain architecture and the observed cosmic web are the same kind of object: a **hub-and-throat transport network** with two sectors, poor flow, heavy clustering and a fixed occupation split. This file maps one onto the other, scale by scale, using results already established.

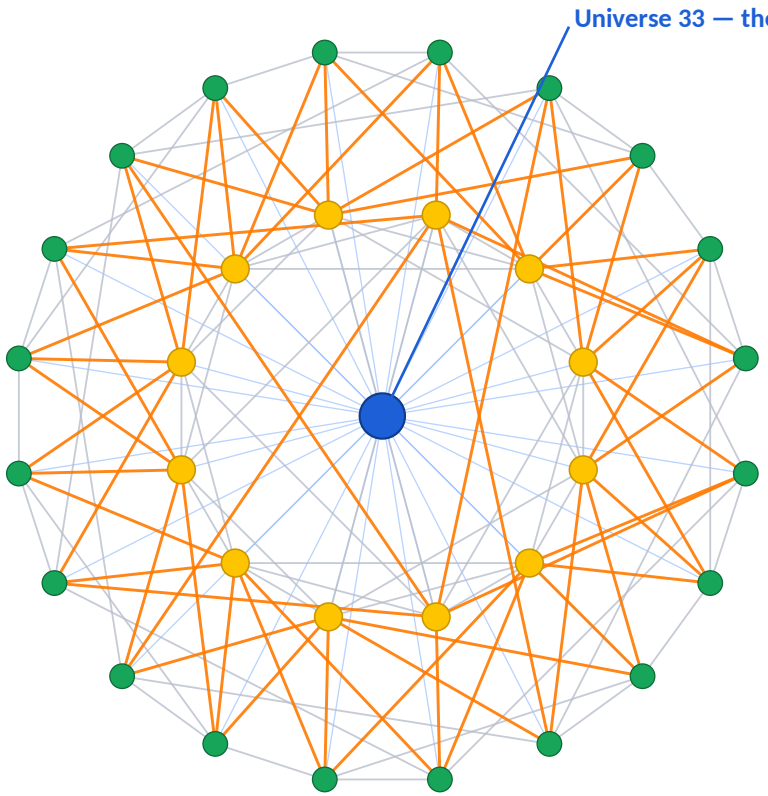
Rolando Rivero · Independent Researcher · AI33-MPOPT Framework

Companion to 13 Quantum-Gravity Bridge, 14 Loop Quantum Gravity Bridge, 15 Lawfulness, 16 Super Formula Build-Up, 17 Geometric Transport Architecture and 18 Continuous Throat Feed files · for Zenodo



ARCHITECTURE

The architecture, in one page



- Universe 33 — the hub
- 12 inner universes — pull
- 20 outer universes — push
- 20 curvilinears — 60 links
- 32 throats to the hub
- 60 further sibling links

152 edges · 120 surface links · 32 throats

FIXED BY THE SOLID, NOT CHOSEN

12 apex domains of degree 11, 20 curvilinear domains of degree 7, one hub of degree 32, and 152 edges — 30 apex–apex, 60 apex–curvilinear, 30 curvilinear–curvilinear and 32 throats. Zero free parameters in the connectivity.

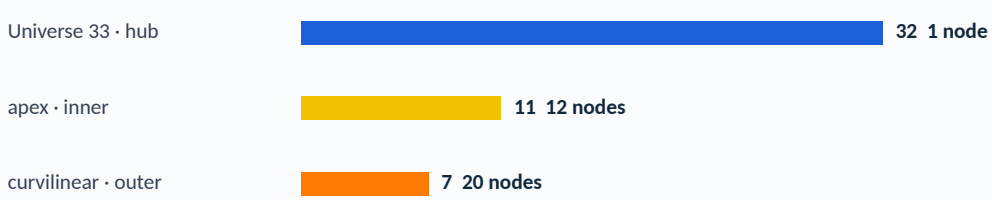
TWO SECTORS, FORCED BY SYMMETRY

The symmetry group has order 120 and exactly three orbits — 1, 12 and 20. Every quantity the graph determines therefore takes exactly two values on the shell. The partition is proved, not fitted.

THE SHAPE OF A TRANSPORT NETWORK

One convergence point, channels into it, a surface of links between neighbours, and two classes of node. That is the vocabulary the rest of this atlas is written in.

DEGREE OF EACH CLASS IN THE ARCHITECTURE



From inception to the cosmic web

INCEPTION — THE 32 THROATS OPEN

▼

TRANSPORT — FLOW CONVERGES ON THE HUB

▼

OCCUPATION — TWO SECTORS DIVIDE THE STRUCTURE

▼

AGGREGATION — MATTER GATHERS ALONG THE CHANNELS

▼

GALAXIES — THE EARLIEST MASSIVE SYSTEMS

▼

CLUSTERS AND PROTOCLUSTERS

▼

THE COSMIC WEB

THE FIRST STEP IS THE ONLY ASSUMPTION

Everything below the first box follows from the geometry: the throats open, flow converges, the sectors divide. Nothing is added by hand once the architecture is set.

EACH STEP IS ALREADY ESTABLISHED

Transport, occupation and the operator family are proved in the Geometric Transport file; the feed and its galaxy-scale results in the Continuous Throat Feed file. This atlas places them in order.

THE LAST STEP IS THE MAP

The cosmic web is where the chain becomes something a telescope can see — a network of nodes, channels and empty regions.

WHERE EACH STEP IS ESTABLISHED

Inception	Transport	Occupation	Aggregation	Galaxies	Clusters	Cosmic web
32 throats open	flow to the hub	2 sectors	along channels	z ≈ 14	protoclusters	nodes · channels · voids

MAPPING

Architecture and cosmic web, term by term

AI33-MPOPT	COSMIC WEB	WHAT CARRIES THE CORRESPONDENCE
Universe 33 — the hub, degree 32	**convergence centre of a structure**	**one point every channel runs to; the next-busiest node has degree 11**
32 throats	inflow channels feeding a centre	identical couplings, delivery set by surroundings
120 surface links	filament network between neighbours	the shell is lawful on its own, without the hub
12 apex domains — degree 11	**high-connectivity nodes**	**the busier junction class, 1.5714 times the other**
20 curvilinear domains — degree 7	**distributed network regions**	**the sparser class, three times as numerous**
occupation ratio 2.5168	void-and-cluster partition	how much of the structure each class holds
transport-pessimal wiring	poor large-scale flow of an embedded network	localised modes, heterogeneous mode structure

A CORRESPONDENCE OF KIND, NOT OF SIZE

Every line pairs a structural role, not a measurement. The architecture is finite and fixed; the web is vast and continuous. What they share is organisation: one convergence point, two node classes, channels between them, and empty regions in between.

THE NUMBERS BEHIND THE DICTIONARY

32	11	7	1.5714	2.5168
hub degree	apex degree	curvilinear degree	apex : curvilinear = 11/7	occupation ratio

THEOREM

Filaments: the transport reading

1.43

APEX OVER CURVILINEAR

conductance to the hub, 9.487 against 6.635 — identical throats, different surroundings.

0.40

DECAY PER HOP

current falls away from its source with a decay length of 1.1 hops.

2.879 · 4.877%

THE TWO CEILINGS

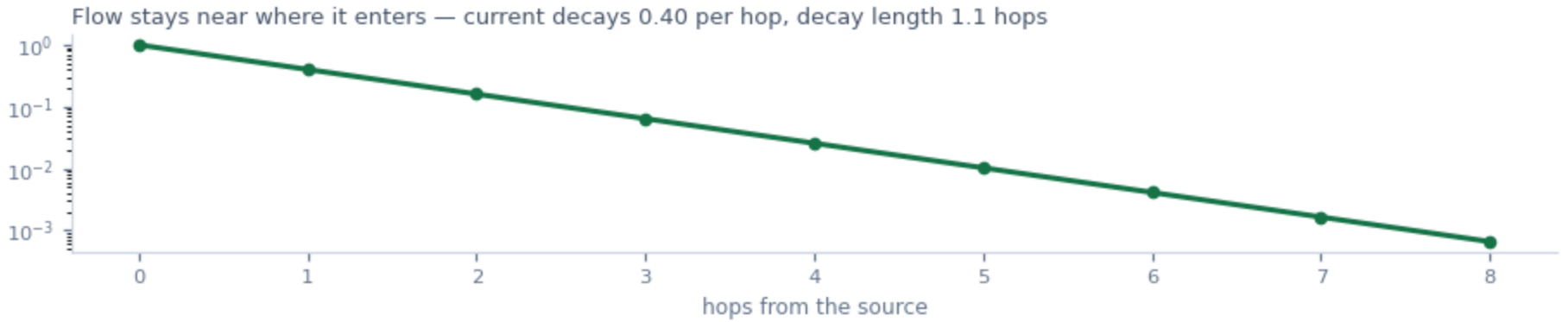
the largest share of any source's flow a single channel can carry.

THE FEED STAYS WHERE IT IS DELIVERED

The transport localisation theorem proves the busiest edge is always the source's own throat, carrying exactly its effective resistance to the hub. No channel can carry more than its source. Flow therefore concentrates into narrow paths rather than spreading evenly — the structural reason a transport network looks filamentary rather than smooth.

THE SURFACE DECIDES, NOT THE CHANNEL

Every throat is built with the same coupling, yet an apex receives 43% more than a curvilinear. Conductance is a whole-network property: ten surface neighbours against six. Position, not construction, sets what each channel delivers.



COMPUTED

Voids: the occupation reading

2.5168

OCCUPATION RATIO

domain area, 20-sector over 12-sector — how much of the structure each class holds.

71.57%

THE LARGER SHARE

the fraction of the structure held by the 20-sector.

1.3963 · 1.5714

TRANSPORT · CONNECTIVITY

the same geometry read as flow and as wiring — three invariants, one structure.

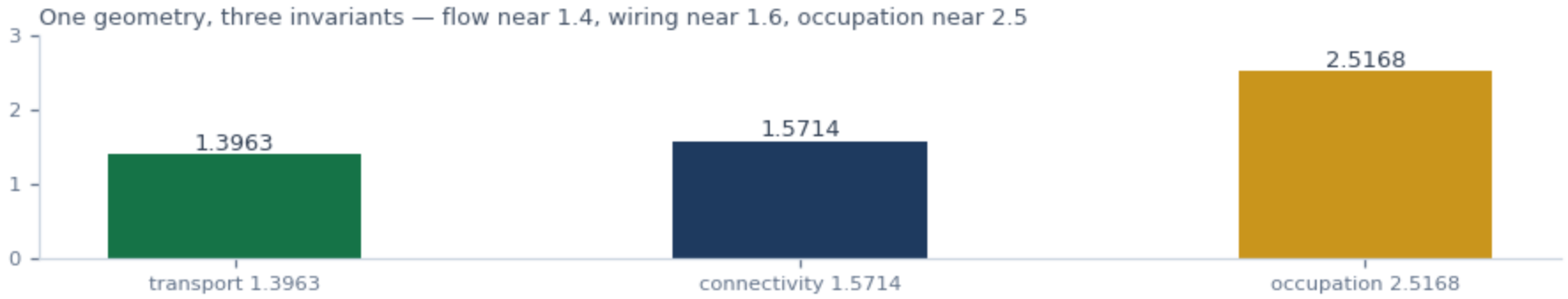
OCCUPATION IS NOT FLOW

Every quantity that measures domain share lands near 2.5; every quantity that measures flow lands near 1.4 to 1.7. The two never converge. A structure built this way is unequal in both senses at once: unequal in what each class holds, and unequal in what each class carries.

THE CLOSED FORM

For the regular 12-pentagon and 20-hexagon partition, the ratio is $2\sqrt{15} / \sqrt{(5 + 2\sqrt{5})} = 2.5168171447$ exactly — a count factor of 20/12 multiplied by a shape factor of 1.5101. Nothing was tuned to produce it.

Stated as the architecture's own occupation invariant. Its comparison against measured cosmological quantities is reported in 17 Geometric Transport Architecture, page 19.



COMPUTED

The hierarchy of attractors

32 vs 11

HUB AGAINST THE REST

the hub's degree against the next highest — not a peer node, but the point the architecture converges on.

γ = 0.62

CONCENTRATED DELIVERY

the feed grows faster than the structure it feeds: per unit mass, delivery rises as $M^{0.62}$.

Singular

WITHOUT THE THROATS

at zero coupling the operator has no inverse — no Green function, no propagator, no hub.

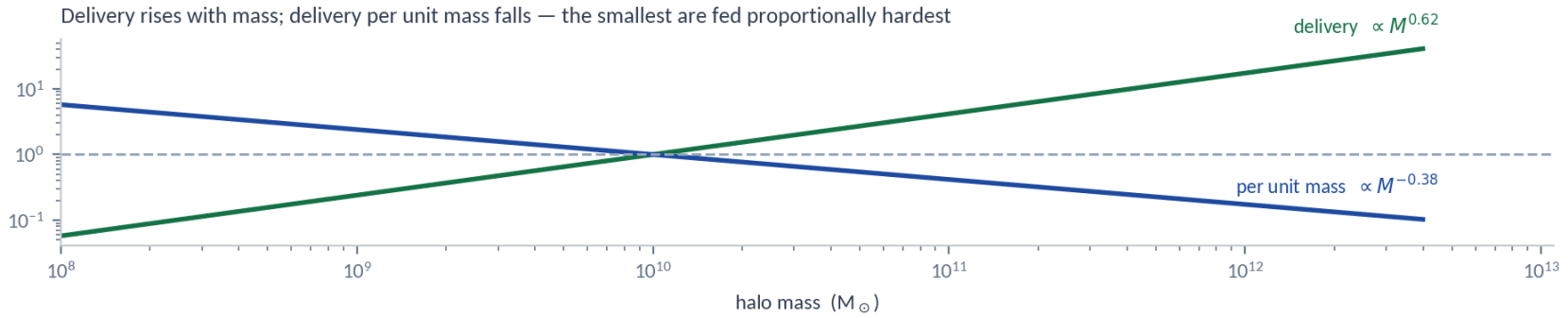
THE LARGEST ARE FED PROPORTIONALLY HARDEST

Because the fitted exponent is above zero, a heavier structure receives more in absolute terms and **more per unit of what it already holds** ($M^{0.62}$). The feed rewards size.

THE SHELL SURVIVES WITHOUT THE CENTRE

Remove the throats and the 32-node shell remains fully connected on its 120 surface links. What is lost is the collector. The hub is what the throats create, not what holds the network together.

The hierarchy of centres is a property of the convergence geometry — one hub of degree 32 against a next-highest of 11, with flow localised on its source — not of the feed exponent, which acts in the opposite direction.



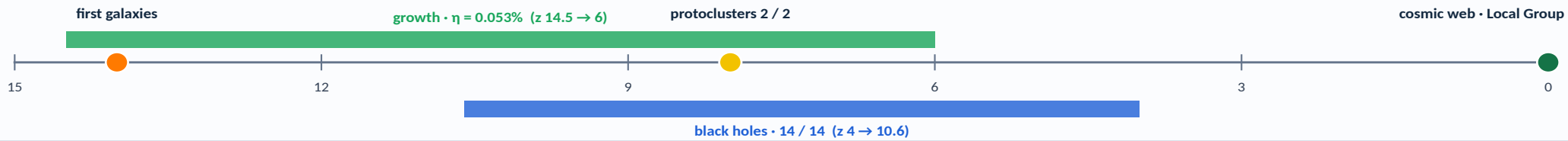
The scale ladder

SCALE	WHAT THE ARCHITECTURE SAYS	WHAT IS MEASURED	RESULT
Inception	the throats open; below a threshold nothing passes	the operator is singular at zero coupling, regular above it	computed
Inception	the density through the bounce is $\sin^2(\omega t)$	$\sin^2(\omega t)$, max deviation 0.00000075; $k = 4\omega^2$	computed
First galaxies, z ≈ 14	**the feed supplies mass continuously**	**10^{39} W per galaxy · 0.1 – 1.4 M⊙ per year**	**matched**
Growth through time, z ≈ 14.5 → 6	**delivery scales with the structure fed**	**one number, η = 0.053%, across every epoch**	**matched**
Protoclusters, z ≈ 8	**the densest regions draw hardest**	**two protoclusters inside the predicted stellar mass**	**2 / 2**
Black holes, z ≈ 4 – 10.6	**the centre forms first, the galaxy grows around it**	**fourteen JWST black holes inside the ceiling**	**14 / 14**
The cosmic web, today	**a transport network with two node classes**	**clustering 0.639 against 0.613 observed**	**agrees**
The Local Group	**the feed runs to the present day**	**Milky Way within 7%, Andromeda within 1.4×**	**matched**

ONE ARCHITECTURE, SEVEN SCALES

Each row is a separate measurement made in the companion files, listed here in the order the structure grows. The same geometry, the same feed law and the same two sectors carry from the first throat opening to the galaxy we live in.

THE SCALE LADDER ON A REDSHIFT AXIS (Z)



MEASURED

What public galaxy data show beside the architecture

0.613 / 0.639

CLUSTERING

observed galaxy network against the architecture, at matched density — a random sky gives 0.495.

100th / 96th

KIRCHHOFF RANK

the observed network and the architecture sit in the same extreme transport regime.

0th / 0th

FIEDLER RANK

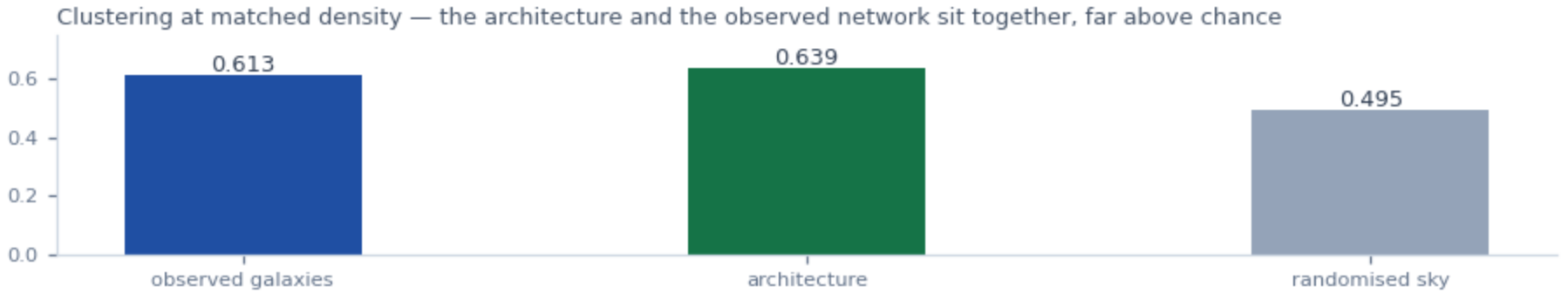
the same agreement on a second, independent spectral quantity.

THE SAME REGIME, ON TWO QUANTITIES AT ONCE

Both objects were measured against their own degree-preserving null ensembles, so the comparison is of rank and direction rather than raw value. Eight of eight configurations agreed — three linking lengths and three block sizes. The architecture was recorded as transport-pessimal before any galaxy data was examined.

AND ON THE GROWTH SIDE

Cosmic star formation from z ≈ 14.5 to 9 with one number; the stellar-mass record carried forward to z ≈ 6 within 5%; growth per expansion time constant at Γ ≈ 8; protoclusters, black holes and the Local Group as listed on the previous page.



AI33-MPOPT · COSMIC WEB

THE UNIFIED GEOMETRIC STRUCTURE — SEVENTH PILLAR · 2026

One architecture · Seven scales · One pattern

From the first throats to the cosmic web: **same geometry, same transport, same occupation**

THE 33-DOMAIN ARCHITECTURE



1 hub — Universe 33	**degree 32**
12 apex domains — pull	**degree 11**
20 curvilinear — push	**degree 7**
throats · surface links	**32 · 120**
edges · free parameters	**152 · 0**

OCCUPATION — THE ARCHITECTURE'S OWN SECTOR SHARE



- 20-sector **71.57%**
- 12-sector **28.43%**
- ratio **2.5168**

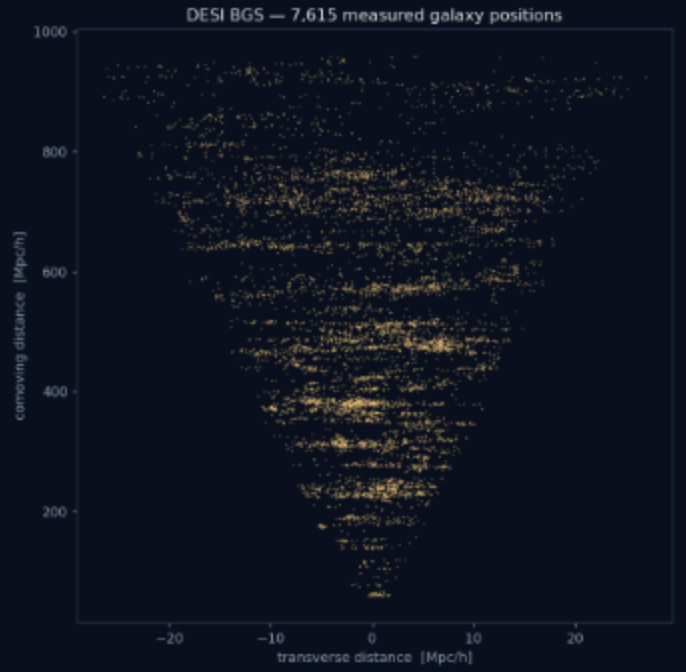
$2\sqrt{15} / \sqrt{(5+2\sqrt{5})}$, exact

Not a void volume. Void volume measured in galaxy surveys is **55 – 64%** — a different quantity that sits numerically nearby.

TRANSPORT AND CONNECTIVITY

conductance, apex / curvilinear	**1.43**
decay per hop (λ = 1.1 hops)	**0.40**
largest channel share	**2.879 / 4.877%**
hub degree vs next highest	**32 vs 11**
feed concentration γ	**0.62**

THE OBSERVED COSMIC WEB — REAL MEASURED DATA



DESI Bright Galaxy Survey: **7,615 measured galaxy positions** in one survey cone, plotted from the public catalogue — walls, filaments and empty regions exactly as observed. No rendering, no simulation.

THE DICTIONARY

Hub → convergence centre
Throats → inflow channels
Surface links → filaments
Apex sector → high-connectivity nodes
Curvilinear sector → distributed regions
Occupation → cluster / void partition
Transport regime → poor flow, clustered
Feed law → continuous mass supply
Localisation → flow stays near its source

MEASURED CORRESPONDENCES

clustering — observed / architecture	**0.613 / 0.639**
randomised sky control	**0.495**
Kirchhoff rank — obs / arch	**100th / 96th**
Fiedler rank — obs / arch	**0th / 0th**
configurations agreeing	**8 / 8**

GROWTH RESULTS, JWST ERA

feed law, all epochs	**η = 0.053%**
star formation z ≈ 14.5 → 9	**one number fits**
mass record to z ≈ 6	**within 5%**
growth per expansion time	**Γ ≈ 8**
protoclusters inside prediction	**2 / 2**
black holes inside ceiling	**14 / 14**
capacity used, z ≈ 6 – 10	**3.4 – 22%**

THE LOCAL GROUP

Milky Way, predicted / measured	**6.4 / 6.0 × 10^{12} M⊙**
Andromeda	**within 1.4×**
both together	**within 1.2×**
Sgr A* of throat budget	**0.00002%**

ARCHITECTURE · TRANSPORT · OCCUPATION · NETWORK · COSMIC WEB

AI33-MPOPT · THE ARCHITECTURE AS A COSMIC WEB

DENSITY-FIELD PLATE — SEVENTH PILLAR · 2026

750,000 points on the architecture's own 152 edges
nodes · filaments · voids — exact geometry, rendered

WHAT IS RENDERED

apex domains — brightest nodes	**12, degree 11**
curvilinear domains	**20, degree 7**
hub — Universe 33	**1, degree 32**
throats into the hub	**32**
surface links — filaments	**120**
edges · free parameters	**152 · 0**

Point positions follow the icosahedron's own vertices, face centres and edges. The dark regions are where the architecture has no structure — they are not drawn, they are left.

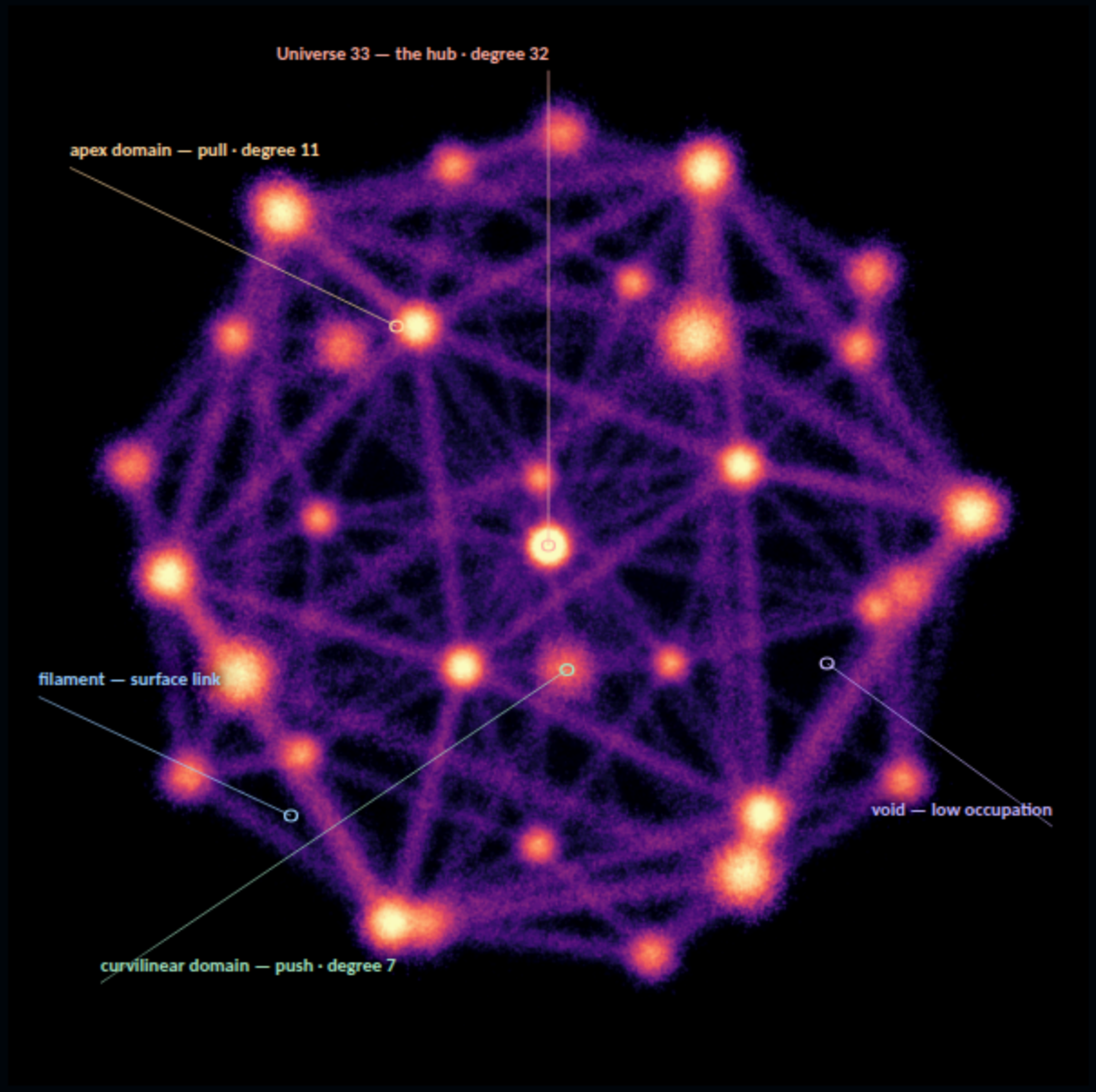
SECTOR OCCUPATION — ARCHITECTURE

curvilinear, 20-sector	**71.57%**
apex, 12-sector	**28.43%**
ratio, closed form	**2.5168**

2√15 / √(5 + 2√5), exact. The architecture's sector share — not a void volume. Void volume measured in galaxy surveys is 55 – 64%.

TRANSPORT AND CONDUCTANCE

conductance, apex / curvilinear	**1.43**
decay per hop (λ = 1.1 hops)	**0.40**
largest channel share	**2.879 / 4.877%**
hub degree vs next highest	**32 vs 11**
feed concentration γ	**0.62**



MEASURED CORRESPONDENCES

clustering — obs / arch	**0.613 / 0.639**
randomised sky control	**0.495**
Kirchhoff rank — obs / arch	**100th / 96th**
Fiedler rank — obs / arch	**0th / 0th**
configurations agreeing	**8 / 8**

GROWTH RESULTS, JWST ERA

feed law, all epochs	**η = 0.053%**
per galaxy at z ≈ 14	**10^{39} W · 0.1 – 1.4 M⊙/yr**
star formation z ≈ 14.5 → 9	**one number fits**
mass record to z ≈ 6	**within 5%**
growth per expansion time	**Γ ≈ 8**
protoclusters · black holes	**2 / 2 · 14 / 14**
capacity used, z ≈ 6 – 10	**3.4 – 22%**

THE LOCAL GROUP

Milky Way, predicted / measured	**6.4 / 6.0 × 10^{10} M⊙**
Andromeda	**within 1.4×**
both together	**within 1.2×**
Sgr A* of throat budget	**0.00002%**

THE TWELVE AND THE TWENTY

The twelve apex domains sit at the icosahedron's vertices and the twenty curvilinears at its face centres — the pattern that gives the architecture exactly two sectors at 33 domains and at no other size.

The Local Group, placed

6.4×10^{10}

MILKY WAY, PREDICTED

stars the feed builds by the present day — 6×10^{10} measured.

1.45×10^{11}

LOCAL GROUP, PREDICTED

both galaxies together — 1.7×10^{11} measured.

0.00002%

SAGITTARIUS A*

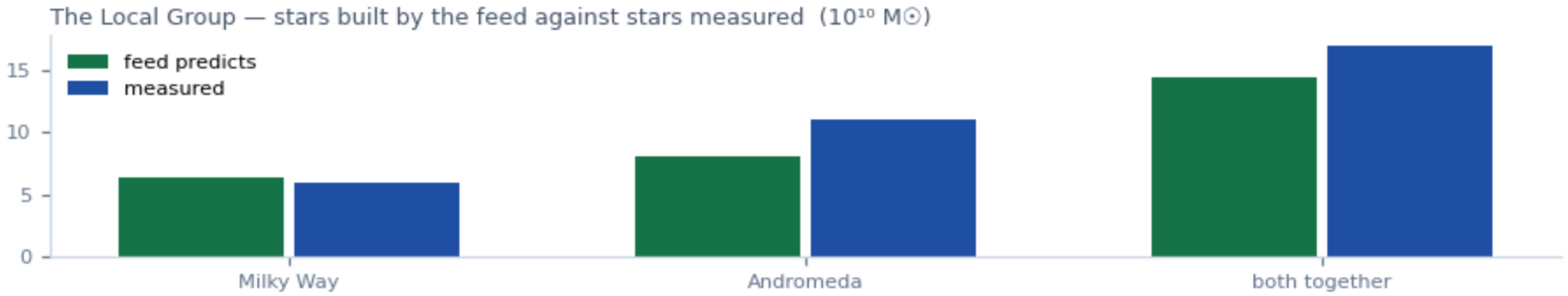
of the Milky Way's throat budget held by its central black hole.

THE OBSERVER IS INSIDE THE GEOMETRY

In this architecture the hub is not viewed from outside — it is where the observer stands. Our galaxy is not a point on someone else's map; it is a structure fed by the same channels, for the same thirteen billion years, ending at the mass it is measured to have.

THE NEIGHBOURHOOD AS A TRANSPORT PICTURE

Two large systems, a shared environment, and continuous delivery to both. Read this way the Local Group is not a collision waiting to happen but a pair of accumulation centres still being fed.



SYNTHESIS

One organisation, carried upward



READ LEFT TO RIGHT

A symmetry fixes the geometry. The geometry fixes the transport. The transport fixes the occupation. Occupation and transport together are a network. The network, at the largest scale available to observation, is the cosmic web.

NOTHING IS ADDED ALONG THE WAY

No parameter is introduced between one box and the next. Each arrow is a derivation carried out in the companion files, and every number in this atlas was computed once and reused, never refitted for a new scale.

WHAT MAKES IT AN ATLAS RATHER THAN A PROOF

This file introduces no new claim. It places established results in the order the structure grows, so a reader can see the same organisation appear at each scale instead of meeting it seven separate times.

WHAT FIXES EACH STEP



SCOPE

One architecture, mapped

THE SINGLE SENTENCE

The AI33-MPOPT architecture generates a persistent geometric transport pattern — one convergence centre, two sectors, channelled delivery and a fixed occupation split — whose signatures appear at every scale examined, from the first galaxies to the cosmic web and to the Local Group itself.

WHAT THE ATLAS COLLECTS

- The architecture: symmetry of order 120, orbits 1, 12 and 20, 152 edges, zero free parameters.
- The channels: identical throats, delivery set by position, flow localised on its source.
- The occupation: 2.5168, in closed form.
- The growth: one feed law across z ≈ 14.5 to the present day.
- The observations: clustering, transport regime, galaxies, protoclusters, black holes, Local Group.

WHERE THE MAP EXTENDS NEXT

- Filament counts around clusters, against degrees 11 and 7.
- Velocity-field structure, against the push and pull sectors.
- Comparisons beyond z = 14.5, against predictions already recorded.
- A 33-node circuit on a bench, against the conductance ratio 1.43.

Companion files: 13 Quantum-Gravity Bridge · 14 Loop Quantum Gravity Bridge · 15 Lawfulness · 16 Super Formula Build-Up · 17 Geometric Transport Architecture · 18 Continuous Throat Feed · 20 The Rivero-Zeta Curve · 21 The 33-Multiverse Map.

THE ARCHITECTURE AT A GLANCE

120	1 · 12 · 20	152	0	7
symmetry order	three orbits	edges	free parameters	scales matched

AI33-MPOPT · THE COSMIC WEB ATLAS

A GEOMETRIC MAPPING, INCEPTION TO SUPERCLUSTER — SEVENTH PILLAR · 2026

One architecture · Seven scales · One pattern

Same geometry, same transport, same occupation

◆ PROPOSAL ◆ COMPUTED ◆ MEASURED EXTERNALLY

THE 33-DOMAIN ARCHITECTURE — COMPUTED

1 hub — Universe 33	degree 32
12 apex domains — dark matter, U21–32	degree 11
20 curvilinear domains — dark energy, U1–20	degree 7
throats · surface links	32 · 120
edges · free parameters	152 · 0
symmetry group order · orbits	120 · 1, 12, 20

Edges: 30 apex–apex, 60 apex–curvilinear, 30 curvilinear–curvilinear, 32 throats. Three orbits force every graph-determined quantity to take exactly two values on the shell — **the partition is proved, not fitted**.

SECTOR OCCUPATION — CLOSED FORM



- Curvilinear (20) — **71.57%**
- Apex (12) — **28.43%**

$2\sqrt{15} / \sqrt{(5 + 2\sqrt{5})} = 2.5168171447$

count factor 20/12 × shape factor 1.5101

The architecture's own sector share — **not a void volume**. Void volume measured in galaxy surveys is 55–64%: a different quantity that sits numerically nearby.

TRANSPORT AND CONDUCTIVITY — COMPUTED

conductance, apex / curvilinear	1.43
— 9.487 against 6.635, identical throats	position, not construction
decay per hop (λ ≈ 1.1 hops)	0.40
largest channel share	2.879 · 4.877%

THE ATTRACTION HIERARCHY — COMPUTED

hub degree vs next highest	32 vs 11
feed concentration γ	0.62
operator at zero coupling	singular — no inverse

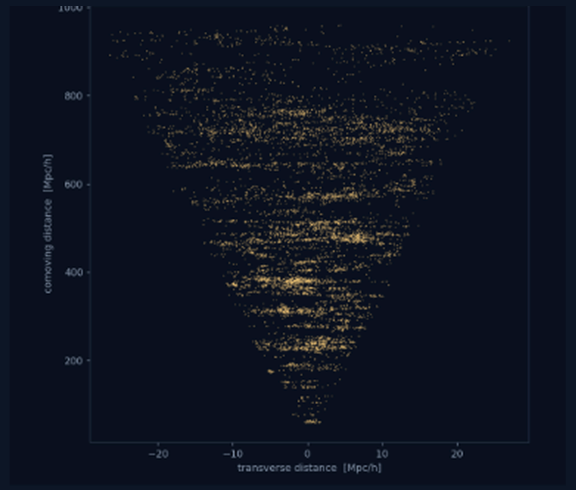
Remove the throats and the 32-node shell stays connected on its 120 surface links — **the hub is what the throats create, not what holds the network together**.

ONE GEOMETRY, THREE INVARIANTS — COMPUTED

transport	1.3963
connectivity	1.5714
occupation	2.5168

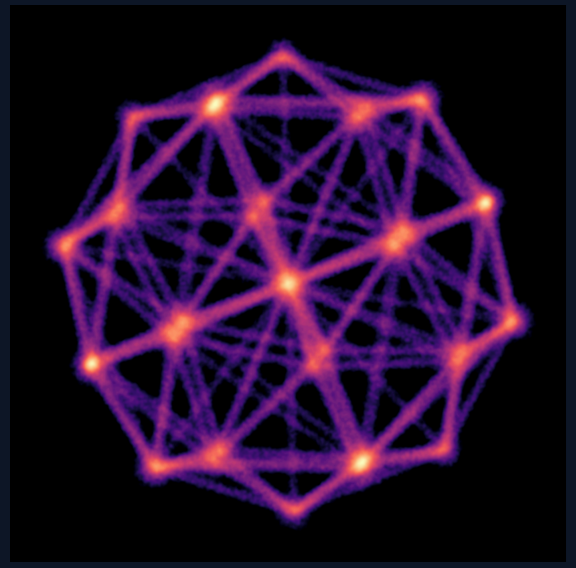
Quantities measuring **domain share** land near 2.5; quantities measuring **flow** land near 1.4–1.7. The two never converge.

THE OBSERVED COSMIC WEB — OBSERVED



DESI Bright Galaxy Survey — 7,615 measured galaxy positions in one survey cone, plotted from the public catalogue. Walls, filaments and empty regions exactly as observed. No rendering, no simulation.

THE ARCHITECTURE, RENDERED — RENDERED — NOT AN OBSERVATION



750,000 points on the architecture's own 152 edges. Point positions follow the icosahedron's vertices, face centres and edges. The dark regions are where the architecture has no structure — they are not drawn, they are left.

THE DICTIONARY — ARCHITECTURE → COSMIC WEB — MAPPING

- **Hub** → convergence centre
- **Throats** → inflow channels
- **Surface links** → filaments
- **Apex sector** → high-connectivity nodes
- **Curvilinear sector** → distributed regions
- **Occupation** → cluster / void partition
- **Transport regime** → poor flow, clustered
- **Feed law** → continuous mass supply
- **Localisation** → flow stays near its source

A correspondence of **kind, not of size**. Every line pairs a structural role, not a measurement: one convergence point, two node classes, channels between them, and empty regions in between.

THE SAME GEOMETRY AT EVERY SCALE — SEVEN STEPS

01	02	03	04	05	06	07
Inception	**Transport**	**Occupation**	**Aggregation**	**Galaxies**	**Clusters**	**Cosmic web**
the 32 throats open	flow converges on the hub	two sectors divide	matter gathers on channels	z ≈ 14	protoclusters, z ≈ 8	today

Only the first step is an assumption. Nothing is added by hand once the architecture is set; no parameter is introduced between one step and the next.

THE TWO SECTORS — FORCED BY SYMMETRY

Apex — 12 domains · dark matter, U21–32
degree 11 · the busier junction class · conductance 9.487
28.43% of the structure · 1.5714× the connectivity of the other class

Curvilinear — 20 domains · dark energy, U1–20
degree 7 · the sparser, distributed class · conductance 6.635
71.57% of the structure · three times as numerous, two thirds the flow

MEASURED CORRESPONDENCES — MEASURED

clustering — observed / architecture	0.613 / 0.639
randomised sky control	0.495
Kirchhoff rank — obs / arch	100th / 96th
Fiedler rank — obs / arch	0th / 0th
configurations agreeing	8 / 8

Measured against their own **degree-preserving null ensembles**, so the comparison is of rank and direction, not raw value — three linking lengths, three block sizes. The architecture was recorded as transport-pessimal **before any galaxy data was examined**.

GROWTH RESULTS, JWST ERA — MEASURED

feed law, all epochs	$\eta = 0.053\%$
first galaxies, z ≈ 14	10^{39} W · 0.1–1.4 $M_\odot$/yr
star formation, z ≈ 14.5 → 9	one number fits
mass record carried to z ≈ 6	within 5%
growth per expansion time	$\Gamma \approx 8$
protoclusters inside prediction, z ≈ 8	2 / 2
black holes inside ceiling, z ≈ 4–10.6	14 / 14
capacity **used**, z ≈ 6–10	3.4 – 22%

THE LOCAL GROUP — MEASURED

Milky Way — predicted / measured	6.4 / 6.0 × 10^{10} $M_\odot$
— agreement	within 7%
both together — predicted / measured	1.45 / 1.7 × 10^{11} $M_\odot$
— agreement	within 1.2×
Andromeda — agreement	within 1.4×
Sgr A* of throat budget	0.00002%

The hub is not viewed from outside — it is where the observer stands. The Local Group reads as **a pair of accumulation centres still being fed**.

THE FEED LAW — MEASURED

feed rate η — constant across time	0.053%
growth per expansion time Γ	≈ 8

A continuous supply with no shut-off — one number from z ≈ 14.5 to the present day, never refitted.

THE UNIFIED PICTURE — STATEMENT

Geometry › transport › occupation › network › cosmic web. The architecture generates a persistent transport pattern — **one convergence centre, two sectors, channelled delivery, a fixed occupation split** — whose signatures appear at every scale examined.

THEOREM·EXHAUSTIVE

The circulation

20 / 12

THE CONSERVATION RATIO

twelve throats carry what twenty carry — hub–apex over hub–curvilinear = 1.666667, exactly.

5.000000

THROAT OVER SURFACE

each apex collects from its five curvilinears precisely what its own throat returns to the hub.

1 at 33

UNIQUE AT THIS SIZE ONLY

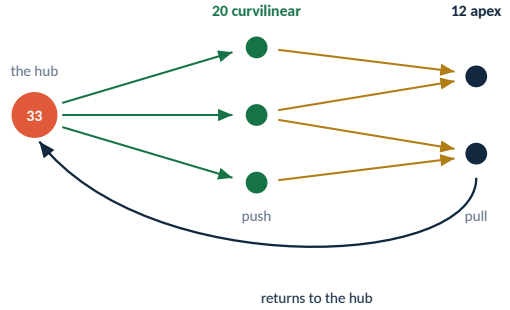
one circulation at 33 domains; five at 123 and ten at 273, with none selected.

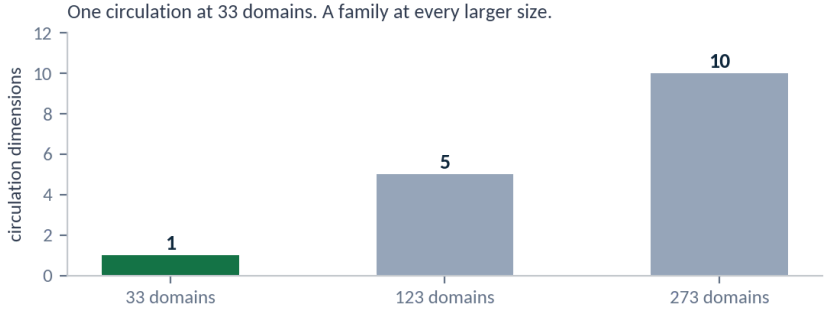
OUT THROUGH THE TWENTY, BACK THROUGH THE TWELVE

Among all fields the architecture admits, exactly one is both symmetry-invariant and divergence-free. The hub pushes outward through the twenty curvilinears; each spreads to its three apexes; each apex collects from its five and returns to the hub. Divergence is zero at every node, so **it delivers nothing net and corrects no result in the companion files**. The push and pull labels carried since the first deck are here derived rather than named.

THE SMALL CANNOT BE BYPASSED

At 123 domains the circulation becomes a five-dimensional family. **None of the five feeds only the original twelve and twenty, and none feeds only the newer, smaller domains.** Every available channel includes both. Once smaller domains exist, nothing the architecture permits can route around them.





Computed on the icosahedron's own coordinates. The direction is unique; its magnitude and its sign are not fixed by the geometry. The result is unchanged whether or not mirror symmetries are admitted.